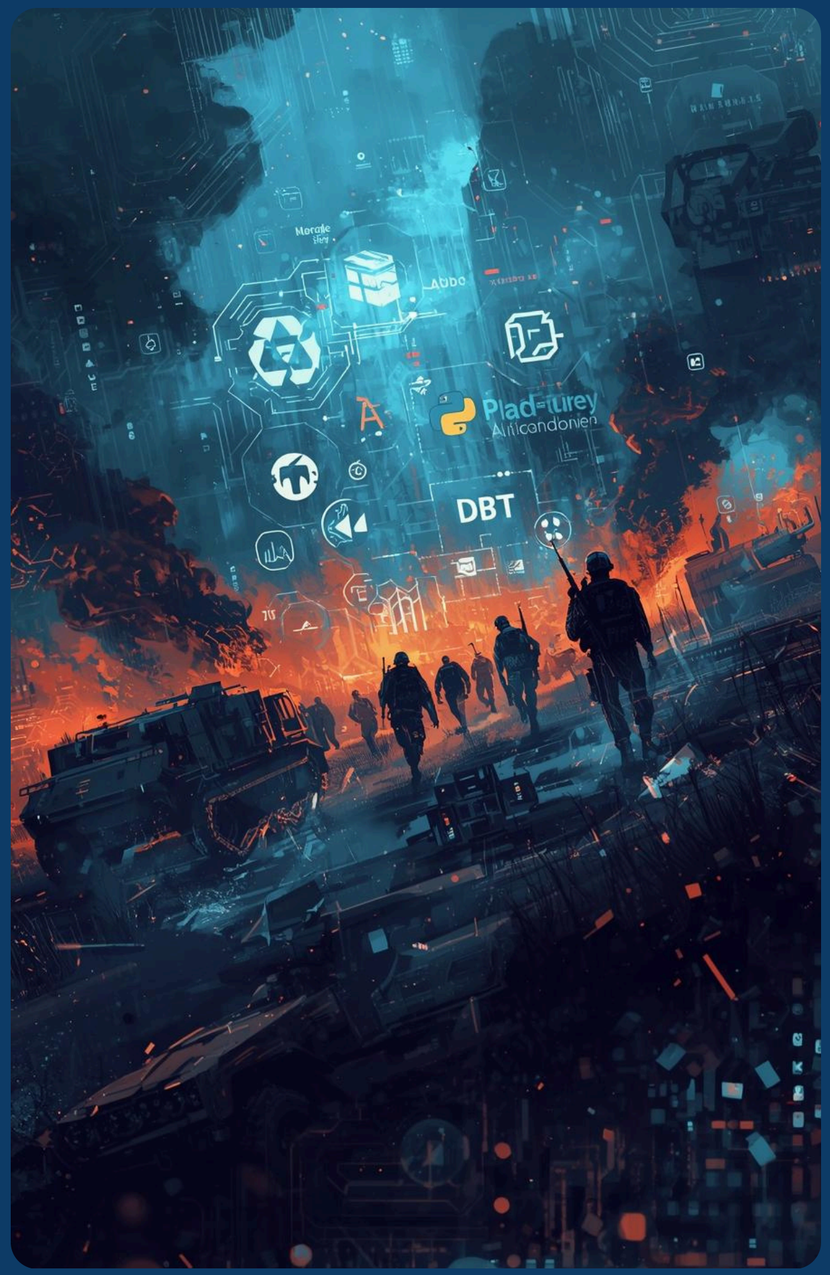
Battlefield 6 Meta Aggregator

Paul Fretté - Python, BigQuery, DBT



Business Challenge

Understanding the gaming landscape today

Gamers and Meta

With over **50 million gamers**, the rapidly evolving meta presents challenges in tracking changes and understanding player preferences, leading to data fragmentation and missed opportunities.

Automation and Insights

Manual analysis has become a bottleneck, requiring automation to harness the vast amount of data effectively and derive strategic insights that can influence gameplay and strategy.

Public Dashboard Opportunity

A public dashboard could provide **real-time insights**, offering players essential metrics and trends that enhance their experience, making informed decisions based on aggregated data.



Technical Architecture Timeline

ELT Pipeline Initiation

The ELT Pipeline was initiated, integrating Python and Selenium for data extraction, setting the stage for robust data management and analytics capabilities in the project.

Data Processing Integration

Data processing was integrated using BigQuery and DBT, significantly improving data transformations while ensuring high-quality insights and supporting the needs of the gamer community.

Streamlit Dashboard Launch

The Streamlit dashboard was successfully launched, providing real-time visualizations and interactive features, empowering users to gain insights from the data and enhancing overall user engagement.

DBT ELT Model

Understanding Data Transformation Levels

The DBT ELT model utilizes **Bronze, Silver, and Gold** tables for effective data transformation, ensuring quality and reliability through 44 data quality tests, enhancing overall data management.

Technical Challenges & Solutions

01

Cloudflare

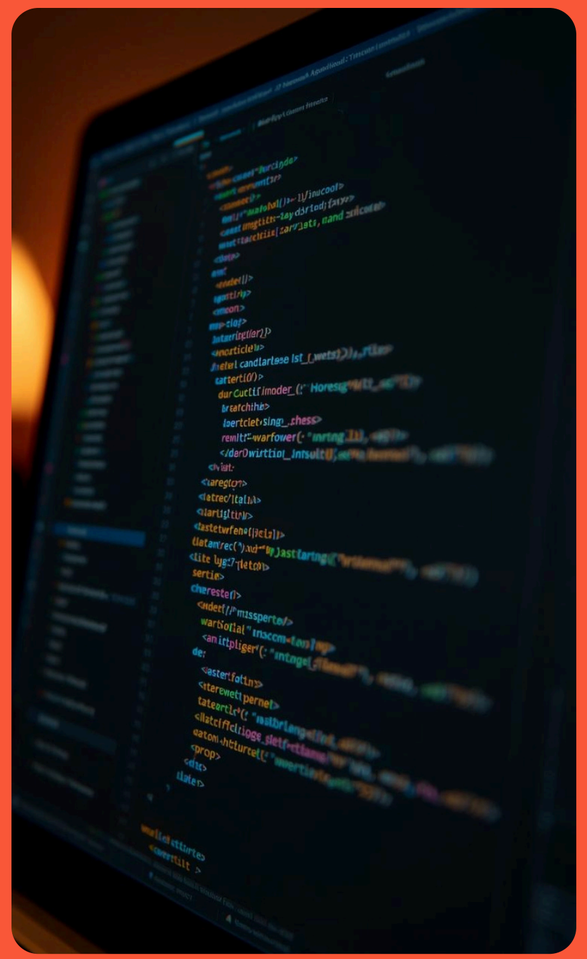
Overcoming **Cloudflare anti-bot blocking** was crucial; we employed Xvfb and an undetected-chromedriver solution to ensure uninterrupted access and achieved a **100% CI/CD success** rate.

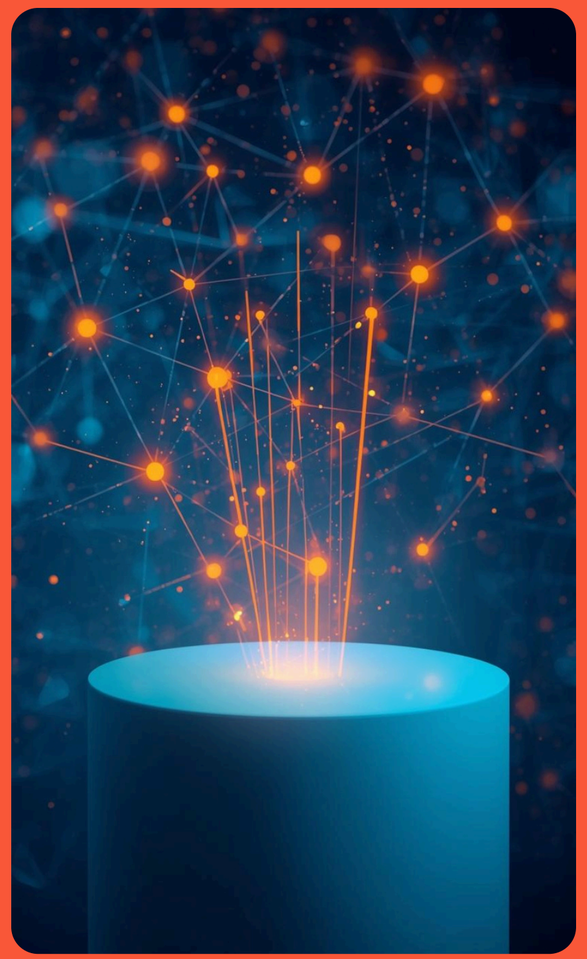
02

Automation

This automation allowed our pipeline to function seamlessly, minimizing human intervention and ensuring consistent data extraction while maintaining **robust efficiency** and reliability throughout the process.

Technical Challenges: Podium Parsing Strategies





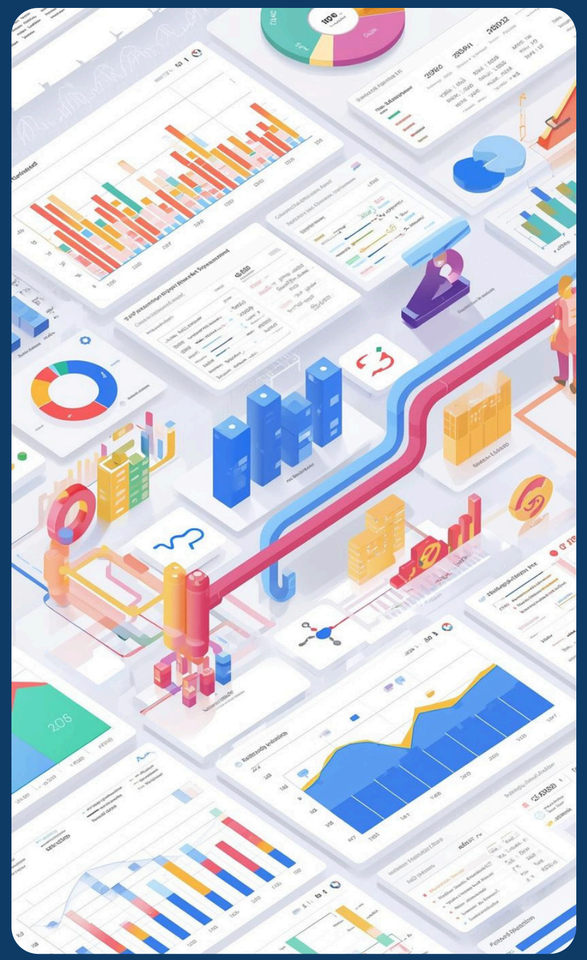
Parsing Strategy

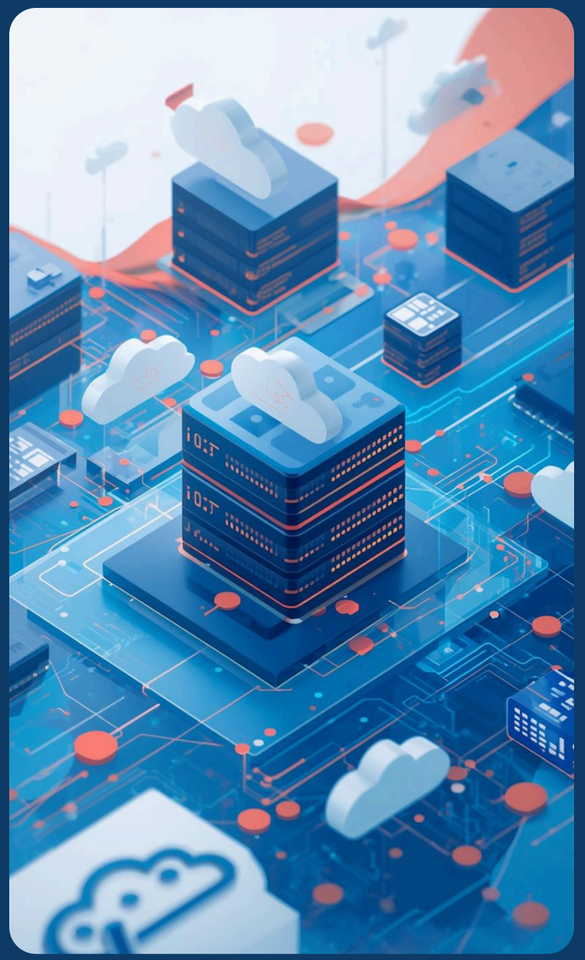
Implemented a **fallback method** for data extraction.

100% Extraction

Achieved a **complete** and accurate dataset from podium.

Addressing BigQuery Data Duplication Challenges





Data Duplication

Implemented sliding window cleaning to reduce redundancy.

Storage Optimization

Achieved -66% storage costs through efficient strategies.

GitHub Artifacts Management



01

Mismatch Issues

Addressing the **mismatch of artifact paths** between CI/CD jobs was crucial for ensuring smooth operation. By aligning upload and download paths, we achieved stable workflows.

02

CI/CD Stability

This alignment led to improved **CI/CD pipeline stability**. The adjustments have minimized errors and disruptions, ensuring that our deployment process runs efficiently and reliably.

Pipeline KPIs

Key Performance Indicators Overview

Success Rate

The pipeline achieved a remarkable **100% success rate**, ensuring reliability and trust in data processing without any errors throughout its operational phases.

Workflow Efficiency

With an average workflow time of only **8 minutes**, the system demonstrates exceptional efficiency, allowing for rapid data analysis and reporting, meeting user demands effectively.

Cost Effectiveness

Operating at a total cost of **$0**, the pipeline illustrates a sustainable approach to data engineering, maximizing resources while minimizing financial impact on the organization.



Streamlit Dashboard Features

Insights into Gameplay and Performance

Our Streamlit dashboard offers an **interactive experience** for users, showcasing essential features like leaderboards, weapon meta analysis, and player deep dives for informed decision-making.







Security Measures

Best Practices for Data Protection

GitHub Secrets

We utilize **GitHub Secrets** to store sensitive information securely, ensuring that access tokens and credentials are protected and not exposed in our code repositories.

Security Audits

Regular **security audits** are conducted to identify vulnerabilities, ensuring our systems are robust against potential threats and aligning with industry best practices for data protection.

Structured Logs

Implementing **structured logs** allows us to efficiently monitor and analyze events and activities within our systems, enhancing our ability to detect and respond to security incidents proactively.



Scalability Roadmap

Multi-Game Support

Implementation of multi-game support will allow our platform to cater to a wider audience, enhancing user engagement and expanding our reach within the gaming community.

ML Integration

Integration of machine learning capabilities will enable automated insights and recommendations, further improving user experience and offering personalized content based on player behavior and preferences.

Public API Launch

The launch of a public API will facilitate third-party integrations, enabling developers to create applications and services that leverage our data for improved gaming experiences and analytics.

Mobile App Development

Development of a mobile app will provide users with on-the-go access to our dashboard and features, increasing accessibility and convenience for gamers and data enthusiasts alike.

Lessons Learned

Key Insights from the Project

Scraping Techniques

Mastering **web scraping** enabled us to gather critical gaming data efficiently, allowing for rapid analysis and decision-making that directly impacted our project's success and agility.

ELT Process

Adopting the **ELT approach** facilitated smoother data transformations and improved data quality, ensuring the pipeline's reliability and enhancing our ability to derive valuable insights for users.

CI/CD Implementation

Implementing a **CI/CD pipeline** streamlined our development process, reducing deployment times and increasing code reliability, which significantly contributed to our overall project stability and effectiveness.



Open Source Community Engagement

Foster Collaboration and Learning Together





Join our **open-source initiative** to democratize gaming data. Collaborate, contribute, and share your ideas to enhance the project while fostering a vibrant learning community.



Thank You

Contact Us for More Information